WSDB Introduction into PSQL client

One of the ways to access the database is through the psql client. If it is not installed in your system, you need to install it. In Ubuntu, there is a postgresql-client-common package. In MacOSX please check [1].

While in order to run the queries any psql version would do, but it's better to install psql version 12 or later, as that will allow you to run information commands like \d

In order to connect to the database using psql you need to use the following command

```
psql -U your_login -h cappc127.ast.cam.ac.uk wsdb [-p 5432]
```

```
psql -U your_login -h cappc127.ast.cam.ac.uk -d wsdb [-p 5432]
```

If you don't want to type this all the time you can set the environment variables

```
export PGDATABASE=wsdb
export PGUSER=your_login
export PGHOST=cappc127
```

If you don't want to type the password all the time, you can create the .pgpass file in your home directory and put in there

```
cappc127:5432:wsdb:your_login:your_password
```

IMPORTANT The file must not be readable by anyone else (chmod og-r) Then you can just call psql without any arguments:

```
psql
```

In psql there are several main important commands:

- \d -- it shows you the list of your tables.

E.g.:

```
wsdb=> \d
          List of relations
 Schema  |    Name    | Type  |  Owner
---------+------------+-------+---------
 koposov | photometry | table | koposov
 koposov | sources    | table | koposov
```

It also allows you to see what are the columns in the given table and what indexes were created on the table.

```
wsdb=> \d sdssdr7.phototag
      Table "sdssdr7.phototag"
   Column     |   Type   | Modifiers
--------------+----------+-----------
 objid        | bigint   |
 skyversion   | smallint |
 run          | smallint |
 rerun        | smallint |
.........
size          | real     |
Indexes:
    "sdssdr7_phototag_objid_idx" btree (objid) WITH (fillfactor=100)
    "sdssdr7_phototag_q3c_idx" btree (q3c_ang2ipix(ra, "dec")) WITH (fillfactor=100) CLUSTER
```

- \dn -- it shows the list of namespaces (also called schemas), which will basically show you the list of available catalogs
- \password -- allows you to change your database password

If you want to execute the query in the database and put the result of that query into some file, then you should do:

```
psql -h cappc127 -U your_login -c 'your_query' wsdb > some_outputfile
```

The useful command line switches which you may want to use in that case are '-t', which force psql to print the data only (no header, no ending), '-A' which disable the aligning of columns, and '-F' which allows you to select the delimiter used by psql to delimit columns.

But if the results of your query are really large(multiple Gb), the best way to output the results of it into the file may be

```
psql -U your_login -c "copy ( here_is_your_query_in_brackets ) to stdin with delimiter ' '" -h cappc127 wsdb > some_outputfile
```

If you want to dump the contents of some table (and that table is large), the way to do it is:

```
psql -h cappc127 -U your_login -c "COPY sometable TO STDIN WITH DELIMITER '|' " wsdb > some_outputfile
```

Or If you want CSV (comma separated values) output

```
psql -h cappc127 -U your_login -c "COPY sometable TO STDIN WITH CSV header " wsdb > some_outputfile.csv
```

If you want to populate your own table with some data, you should first create your table with the query like that:

```
CREATE TABLE mytable ( a int, b rel, c double precision, d varchar);
```

Then you can populate the table, if you have your data in some delimiter separated format (e.g. "|"):

```
cat your_file_with_data |  psql  -h cappc127 -U your_login -c "COPY mytable FROM stdin WITH delimiter '|'" wsdb
```

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